

Curriculum Vitae

CONTACT INFORMATION

Yashdeep Singh
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RESEARCH SUMMARY

My research focuses on **popular matching problems** in graphs and hypergraphs. I have shown NP-hardness results for 3-uniform 3-partite hypergraphs, proposed approximation algorithms with quadratic bounds on unpopularity factors, and designed constant-factor approximation algorithms for subcubic and cactus graphs. My work combines algorithm design, approximation, and complexity analysis, aiming to advance the theoretical foundations of matching problems.

RESEARCH INTERESTS

Graph algorithms, popular matchings, hypergraph theory, approximation algorithms, computational complexity.

EDUCATION

Indian Institute of Technology Guwahati, Guwahati, India

Ph.D., Computer Science and Engineering, July 2018 - Feb 2026(Expected)
(thesis submitted)

- Thesis Title: *Popular matching problem in special cases of hypergraphs and non-bipartite graphs.*
- Supervisor: Prof. Sushanta Karmakar

M.Tech., Computer Science and Engineering, **7.87** CPI, July 2013 - June 2015

- Thesis Title: *Data Delivery by Energy-Constrained Mobile Agents.*
- Supervisor: Prof. Sushanta Karmakar

Uttar Pradesh Technical University, Lucknow, UP, India

B.Tech., Computer Science and Engineering, **67.22%**, July 2009 - June 2013

EXPERIENCE

- **Teaching Assistant (during PhD)**, IIT Guwahati July 2018 – June 2023
 - Teaching assistant for C programming (theory and lab), Algorithm Design (theory and lab), and Randomized Algorithms.
 - Led tutorials and supported course administration.
- **Assistant Professor**, Jaypee University of Information Technology, Waknaghat, HP Aug 2016 – July 2018
 - Taught courses: Introduction to Algorithms, Database Management System, Theory of Computation, Algorithm Design.
 - Developed course material and assignments, and supervised student projects.
 - Mentored undergraduate students and contributed to departmental activities.
 - Organizing Committee Member, International Conference on Parallel, Distributed and Grid Computing (PDGC), 2016 (held at JUIT).
- **Software Engineer**, Collabera Technologies Nov-2015 to July-2016
 - Developed web-based application on ExtJS, PHP, and Oracle.
- **Teaching Assistant (during M.Tech)**, IIT Guwahati July-2013 to June-2015
 - Teaching assistant for C programming lab and Algorithm Design lab.

PUBLICATIONS

Yashdeep Singh and Sushanta Karmakar. An Algorithm for Matching in a (3,3)-Hypergraph With One-Sided Preferences Having Quadratic Unpopularity Factor. *Concurrency and Computation: Practice and Experience*, 37(25-26), e70331 (2025). <https://doi.org/10.1002/cpe.70331>

Yashdeep Singh and Sushanta Karmakar. Improved approximation for unpopularity in (3, 3)-hypergraph matching with one-sided preferences. In *Combinatorial Algorithms - 36th International Workshop, IWOCA 2025, Bozeman, MT, USA, July 21-24, 2025, Proceedings*, volume 15885 of *Lecture Notes in Computer Science*, pages 201–213. Springer, 2025

Yashdeep Singh and Sushanta Karmakar. Bounded unpopularity matching in a (3, 3)-hypergraph with one-sided preferences. In *Twelfth International Symposium on Computing and Networking, CANDAR 2024 - Workshops, Naha, Japan, November 26-29, 2024*, pages 204–210. IEEE, 2024

Yashdeep Singh and Sushanta Karmakar. The popular matching problem in a (3, 3)-hypergraph with one-sided preferences. In *Eleventh International Symposium on Computing and Networking, CANDAR 2023 - Workshops, Matsue, Japan, November 27-30, 2023*, pages 121–127. IEEE, 2023

Yashdeep Singh and Sushanta Karmakar. Popular matching in a 3-uniform 3-partite hypergraph. In *ICDCN '22: 23rd International Conference on Distributed Computing and Networking, Delhi, India, January 4 - 7, 2022*, pages 248–249. ACM, 2022

Yashdeep Singh, and Sushanta Karmakar “Popular Matching in a 3-Uniform 3-Partite Hypergraph with One-sided Preferences: Hardness and Approximation” *International Journal of Foundations of Computer Science*. 2025 (under review)

Yashdeep Singh, and Sushanta Karmakar “Constant unpopularity factor matching in subcubic and cactus graphs” *Discrete Applied Mathematics* (to be submitted)

TECHNICAL SKILLS

Programming Languages: C, Java
Database: MySQL, Oracle
Front-end: Extjs

ACHIEVEMENTS

- Qualified for JRF and Assistant Professor in UGC NET 2017.
- All India Rank 827 (99.6 percentile) in GATE 2013 in CSE.